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MILITARY SPECIFICATION

MIXING PLANT, ASPHALT, BATCH TYPE, PORTABLE, ELECTRIC-MOTOR-DRIVEN, 50 TON PER HOUR CAPACITY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 2,000 pound (lb) batch-mix, manual operated, portable, high-type asphalt mixing plant of 50 tons per hour (tph) minimum capacity. The mixing plant shall be comprised of towable main units with the supplemental component units and accessories to be transported by truck or trailer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-119	- Varnish, Spar, Phenolic-Resin.
TT-V-121	- Varnish, Spar, Water-Resisting.
VV-F-800	- Fuel Oil, Diesel.
VV-L-751	- Lubricating Oil: Chain, Wire-Rope and Exposed Gear.
MMM-A-260	- Adhesive, Water-Resistant (For Sealing Waterproofed Paper).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).
MIL-M-52182 - Melter, Asphalt.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED STD No. 595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provision for Lifting and Tying Down Military Equipment.
MIL-STD-633 - Mobile Electric Power Engine Generator Standard Family Characteristics Data Sheet.
MIL-STD-794 - Parts and Equipment Procedures for Packaging and Packing of.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B15.1 - Mechanical Power - Transmission Apparatus.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C27 - Classification of Fireclay and High-Alumina Refractory Brick.
C213 - Specification for Alumina-Silica-Base Castable Refractories for Boiler Service.
D5 - Test for Penetration of Bituminous Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The asphalt mixing-plant, hereinafter called the plant, shall be an electric-motor-driven, portable type. The plant shall consist of all units, components, and accessories to provide a complete plant capable of continuously mixing not less than 2,000 lb per batch of high quality asphaltic mix at a capacity of not less than 50 tph. The plant shall include, but not be limited to, the following basic units:

a. Mixing and weighing unit consisting of screens, hot storage bins, weighing box, and mixing unit with manual control station.
b. Cold aggregate storage bin consisting of at least four compartments with electric-motor-driven beld feeder(s).
c. One cold-aggregate elevator to transfer cold aggregate from the bin unit into the drier.
d. One aggregate drier of the single-rotating-drum type with automatic temperature-fuel regulating control.

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e. One fines collector of the centrifugal cyclone type with connecting duct sections and exhaust system.
f. One hot-elevator of the totally enclosed, vertical bucket type.
g. One fine feeder with supply bin and elevating mechanism to meter granular mineral filler into the weigh box.
h. Heating system, hot oil, with pumps, regulators, and piping.
i. Burner, low pressure, with pump and blower.
j. Melter, asphalt, hot oil heated, capable of melting and removing asphalt from commercial 55-gallon drums, with a capacity commensurate with mixing plant requirements with not less than 5,000 gallons of heated storage conforming to MIL-M-52181.
k. One wheel mounted asphalt storage tank 10,000 gallon capacity, heated, insulated, with automatic temperature control valves.
l. One wheel mounted fuel storage tank, 5,000 gallon capacity.
m. One complete set of insulated piping, asphalt, for connecting storage tank, melter, transfer pump, and mixing unit.
n. One complete set of insulted piping, hot oil, consisting of manifolds, feeder lines, and connections required for connection to hot oil heater, storage tank, mixing unit, and melter.
o. Electrical control panel complete with motor controls, circuit breakers, switches, relays and auxiliary equipment properly wired, and operable as a central electrical control for the plant.
p. Electrical support items consisting of power and control cables of proper size and number to supply power from control panel to motors, controls, and switches. The main component units shall be mounted on either semitrailers, or cart-type trailers, and may be grouped at the discretion of the supplier as long as it gives maximum portability and assures that the plant can be interconnected mechanically and operate at maximum capacity.

* 3.2 First article. When specified (see 6.2), the contractor shall furnish a complete plant for first article inspection and approval (see 4.2.1 and 6.5).

* 3.3 Standard commercial product. The plant shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of

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used or rebuilt products are allowed under this specification unless otherwise specified. All portions of hoppers, chutes, plates, and metal surfaces, subject to abrasive wear caused by aggregate movement, shall be lined or faced with replaceable, abrasive-resistant, steel plates conforming to ASTM Designation A242, or better. All bolt heads subjected to the flow of material shall not extend beyond surfaces exposed to the material.

* 3.5 Construction. Where no specific construction requirements are specified herein, the manufacturer's best commercial practice shall apply. All parts, components, and attachments shall be identical for each plant furnished under any specific contract. No deviations will be accepted without prior written approval of the contracting officer.

3.5.1 Ease of maintenance. Design of the plant shall permit ready accessibility to all parts for maintenance and repair by the use of conventional general-purpose handtools. Replacement and adjustment of components parts and accessories shall be accomplished with minimum drainage requirements and disturbance to other elements on the plant. Ease of maintenance provisions shall incorporate features insuring clearances between adjacent components to facilitate maintenance and servicing operation.

* 3.5.2 Safety. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded or insulated to the extent necessary to eliminate the hazard.

3.6 Capacity and performance. The plant shall be capable of producing not less than 50 tph, based on a 1-minute mix cycle, of completely mixed hot bituminous asphalt. All components of the plant shall also be of sufficient capacity to fully support the mixing plant at the required capacity.

* 3.7 Transportability. Each wheeled component, when towed, shall be capable of travel over smooth hard-surfaced primary highways at speeds up to 20 miles per hour, travel over unimproved roads and hilly cross-country terrain, and negotiate a 90° turn without weaving.

* 3.7.1 Braking ability. On each wheeled component equipped with brakes, the combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in conformance to DoT Motor Carrier Safety Regulation requirements. When actuated, the emergency breakaway system shall produce smooth emergency brake application.

3.7.2 Ground clearance. Exclusive of the running gear, the ground clearance of each unit shall be not less than 10 inches.

3.8 Servicing and adjusting. Prior to delivery of the component units, the contractor shall adjust and service each component unit for travel, including at least the following: adjust brake system, inflate all tires, and completely lubricate the chassis and components.

3.9 Vehicular component units. The cart-type trailer and semitrailer component units shall be as specified herein.

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3.9.1 Intervehicular connections.

3.9.1.1 Truck-tractor and semitrailer. The semitrailer shall be furnished with airbrake standard commercial couplings, for connecting the truck-tractor airbrake system to the semitrailer system. The coupling shall be installed on the front of the semitrailer, symmetrical about the centerline, in accordance with SAE J702. Coupling clearance shall permit connection of the truck-tractor hoses to the semitrailer without interference with the upper fifth-wheel plate or other components. The airbrake couplings shall be protected with spring covers.

3.9.1.2 Truck and cart-type trailer. All intervehicular components shall be installed on the trailer, so that they shall extend between 36 and 42 inches toward the prime mover from the center of the lunette eye. Two dummy couplings shall be furnished and fastened to the tongue for doubling back and holding the lines. The intervehicular line routing, connected and stowed, shall be such that dropping of the tailgate of the truck will not damage any lines.

3.9.2 Chassis frame. The framing members of the component units shall support the weights and maintain alinement under all conditions of transport and of plant operation as specified herein.

3.9.3 Semitrailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a 36-inch diameter lower fifth-wheel and the entire width of the gooseneck. The fifth-wheel plate and kingpin shall be compatible with a fifth-wheel height of 48 inches ±1 inch. Both longitudinal and transverse reinforcing members shall be provided. At the front of the plate, a front coupling ramp shall be furnished. Ramps that are not flush with the front crossmember shall include at least four vertical gussets and shall be provided with holes for water drainage. A 2-inch throat diameter kingpin, conforming to SAE J700, shall be installed not less than 24 inches nor more than 36 inches from the front of the semitrailer. The center of the kingpin shall be within 1/8 inch of the longitudinal centerline of the semitrailer. The swing clearance rearward from the kingpin to any part of the semitrailer below the upper fifth-wheel plate shall be not less than 74 inches.

* 3.9.4 Cart-type trailer towing attachment. A removable towing attachment shall be attached to the cart-type trailer and shall include a lunette conforming to SAE J847. When the cart-type trailer is in the level towing position, the height of the lunette shall be between 32 and 36 inches from the ground line. Two safety chains shall be furnished on each trailer.

3.9.5 Landing jacks and jack legs. Landing jacks shall be provided on semitrailers and jack legs on cart-type trailers for leveling and support each vehicular component unit, except for the hot-aggregate elevator. The landing jacks and jack legs shall be hand or power operated hydraulic units, and shall be permanently installed on the vehicular component unit. The landing jacks and jack legs shall be of such strength, design, and location that coupling and uncoupling the trailer, with a towing vehicle, will not cause excess stresses. All jacks shall be equipped with square metal mud pads having a ground bearing area of not less than 100 square inches.

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3.9.6 Running gear. The axle, or the forward axle on tandem axles, shall be in alinement with the longitudinal centerline of the vehicular component unit so that the two diagonal measurements, from the center of the lunette or kingpin to the outer center of each hub, do not differ in excess of 1/8 inch. The rear axle of tandem axle assemblies shall be in alinement with the front tandem axle so that the parallel measurements between hub centers on each side do not differ in excess of 1/8 inch. Stub axles or through axles may be furnished. The length of axle tubes or through axles may be increased during manufacture, providing the mounting pads are increased at least an equal distance to insure that the structural rigidity is equal to that of the axles. Axle assembly for the component units shall be either of the single or tandem axle type, as applicable for load distribution and weight. All units shall be equipped with suitable size wheels and tires of the same size in accordance with the recommendations of the Tire and Rim Association, to withstand the maximum loads, stresses, travel speeds, and with adequate road clearance. Each single axle shall be designated for a load of not more than 18,000 lb, and shall be compatible with the brake system in accordance with the loadings.

3.9.7 Brake system. Each wheeled component unit shall be equipped with a service and emergency breakaway, waterproof, all air or air over hydraulic brake system, complying with all applicable DoT Motor Carrier Safety Regulations construction and performance requirements. The brake system shall include service and emergency line filters, an air reservoir, and an emergency relay valve. All brake system components shall be located, installed, and protected so that damage from ground obstructions and debris is prevented and that maximum efficiency and ease of maintenance is obtained.

3.10 Electric equipment.

3.10.1 Electric motors. All electric motors shall be 3 phase, single speed, induction type, and connected for 208 volt, alternating current (ac), 60 Hertz operation. All motors shall be totally enclosed nonventilated or totally enclosed fan-cooled machines conforming to NEMA MG1. Motors shall be suitable for the respective applications in an ambient temperature of 125° Fahrenheit (F). All motors, 25 horsepower (hp) and larger, shall be arranged for sequence starting from nonreversing reduced voltage or current motor starters. The remaining motors may be across-the-line start with overload protection and controlled from the motor control center (see 3.10.2).

3.10.2 Motor control center. The motor control center shall be a free standing, outdoor type, combination motor control unit, rated for 208 volts, 3 phase, 3 wire, ac, conforming to NEMA in a type 3S enclosure. Provision shall be made for remote control stations (start-stop pushbuttons) where required. The motor control center, complete with all motor controllers, components, and accessories shall be suitable for continuous operation under conditions as follows:

a. At ambient temperatures from +125°F to +32°F and at any possible relative humidity.
b. Exposure to severe dust conditions.
c. Exposure to salt air.
d. At any altitude from mean sea level (msl) to 5,000 feet above msl.

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The motor control center shall have a manually operated main circuit breaker on the incoming power supply to the control center. The main circuit breaker may be either integral with the control center, or in a separate enclosure and shall have overload, ground fault, and short circuit protection. The main circuit breaker shall have a shunt trip, with remote trip pushbutton(s) located at the operator position(s), for emergency trip operation. The motor control center shall have an instruction plate permanently affixed in a conspicuous location. In addition to instructions for connecting and operating the plant, the instruction plate shall show the power supply requirement for the plant in bold type, as follows:

POWER SUPPLY REQUIRED
208 VOLTS, 3 PHASE, 3 WIRE
INPUT CAPACITY, MAXIMUM _____KVA (_____(KW))
OPERATING LOAD (AT RATED CAPACITY) _____KVA (_____(KW))
STANDBY LOAD (HOT OIL HEATER, ASPHALT MELTERS, PUMPS) _____KVA (_____(KW))

3.10.2.1 Motor controllers. Motor controllers (starters) shall be ac, general purpose, class A, magnetic controllers for single speed, nonreversing, nonjogging, squirrel-cage induction motors, conforming to NEMA ICS. Controllers for motors of 25 hp and larger shall be reduced voltage or current motor controllers. Controllers for motors of less than 25 hp may be full-voltage (across-the-line) controllers. Insofar as consistent with plant operation, the controllers shall be interlocked to provide sequence starting, with the larger motors to be started first, in order to reduce the required power input capacity to a minimum. The starting sequence shall be included in the connecting and operating instructions.

3.10.3 Power and control cables. All power, control and grounding cables, fittings, and accessories required to completely connect the plant for operation shall be furnished in the length required for each circuit. In addition, three sets of power cables (one three-conductor cable or three one-conductor cables each) not less than 100 feet in length shall be furnished to connect the motor control center to the incoming power supply source. One set of power supply cables shall have adequate load capacity to supply the standby loads, such as the hot oil heating system, asphalt melters, and asphalt storage transfer pumps. The other sets of power supply cables shall have adequate load capacity to supply the maximum power input required to operate the complete plant. All power and control cables for the power supply and for interconnecting the plant, shall be heavy-duty portable cables, conforming to NEMA WC3 or WC7, suitable for direct burial or laying on the earth surface. All portable power cables shall be rated for conductor operating temperature of not less than 75° centigrade. The load on any conductor shall not exceed the allowable ampacities of NFPA No. 70 for direct burial conductors. The grounding system for the complete plant shall include the grounding conductors, fittings, ground rods, and clamps. The grounding conductors shall be suitable size and type for the application.

3.11 Drive components.

3.11.1 V-belt drives. V-belt drives shall be in accordance with ANSI B15.1 and shall have provisions for adjustment.

3.11.2 Gears and gearboxes. All gears, except pugmill mixer timing gears, shall be enclosed in gearboxes and shall operate in an oil bath.

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3.12 Cold-aggregate feeder. The cold-aggregate feeder shall be of hopper and frame construction, and shall consist of a two-bin coarse aggregate hopper and a two-bin fine aggregate hopper, with a total capacity of not less than 20 tons, belt-conveyor feeder, all supported by a rigid steel frame. The cold-aggregate feeder shall discharge the materials into the cold-aggregate feeder elevator buckets or conveyor. Bin vibrators shall be provided for each compartment.

3.12.1 Hoppers. The coarse aggregate and fine aggregate hoppers shall be of welded steel construction, each divided into two bins to permit handling of two sizes of aggregate in each hopper (total of four sizes of aggregate). The bin side plates and divider plates shall prevent bridging or clogging of the aggregate. Each bin shall have a wide opening for receiving the amount of aggregate being dumped into each hopper. Hopper and bin extensions shall be provided for both hoppers. Bins and bin extensions shall be provided with reinforcing to prevent damage to the edges while bins are being charged. Each bin shall be provided with an individual, adjustable control gate, or by other suitable means, to regulate the flow of aggregate. The gate shall be provided with an adjustment lever and position locking device when control gate is used for regulating aggregate flow.

3.12.2 Belt-conveyor feeder(s). The belt-conveyor feeder(s) shall consist of a suitable steel frame, conveyor belt, head-end (drive) pulley, foot-end pulley, carrier rolls, and return rolls. Straight carrier rolls shall be mounted under the belt to support the aggregate as it is discharged from the aggregate hoppers. Steel dustpans shall prevent dust and aggregate from falling onto the backside of the conveyor belt. Steel side plates and rubber flashing shall be provided to prevent spillage of aggregate from the sides as it is discharged for the hopper. Pulleys and rollers shall be mounted on steel shafts and support by antifriction or bronze bearings with dustproof seals. Self-alining bearings for head and tail end pulley shafts shall be furnished and shall be provided with labyrinth seals. The belt-conveyor feeder(s) foot-end section shall be provided with a take up adjustment mechanism to obtain belt tension.

3.13 Cold-aggregate elevator. The cold-aggregate elevator shall be electric-motor-driven, inclined bucket, and chain type. Buckets and chain shall be of manufacturer's standard construction as supplied to the commercial market. Means for adjusting and maintaining alinement and tension of the driving mechanism shall be provided. The elevator shall have a capacity of not less than 50 tph.

3.14 Aggregate drier. The aggregate drier shall consist of a single rotating drum with feed or charging chute, oil burner, forced draft blower and motor, fuel pump, accessories and instruments, and all other components necessary for the operation of the drier as specified herein. When drying aggregate at a discharge temperature of 290°F to 310°F, the capacity of drier shall meet the production output rate of the plant with the dried aggregate having a maximum residual surface moisture content of 5 percent.

3.14.1 Drum. The drier drum shall be a single cylinder constructed of heavy-gage steel plate, capable of withstanding the abrasion wear and heat distortion at peak production for long usage without replacement. The drum shall be provided with replaceable lifting flights which shall lift and drop

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the aggregate in a thin veil through the hot gases produced by the burner. Suitable means shall be provided to discharge the dried aggregate into the boot of the hot-aggregate elevator. The drum shall be driven by a positive or friction drive arrangement. All bearings used in the driving mechanism shall be of the antifriction type. The charging end of the drier drum shall include a flanged exhaust gas duct connection, onto which the fines collector duct shall be attached.

3.14.2 Drier burner and fuel system. The drier fuel system shall include burner, fuel piping, fittings, valves, fuel pump, strainers, sediment and moisture bowl, relief valve, pressure gages, and other items which may be required to provide a complete operating drier fuel system.

3.14.2.1 Burner. The drier burner shall be low pressure, air atomizing, and oil burning. The burner shall meet the performance requirements specified herein, when operated with fuel oil conforming to VV-F-800, grade DF-A, DF-1, or DF-2, and jet fuel conforming to MIL-T-5624, grade JP-4 or JP-5. The burner shall have the capacity of drying the wet aggregates to meet the specified production requirement, and shall operate in conjunction with the forced draft blower for efficient combustion.

3.14.2.2 Fuel piping and pipe fittings. Fuel pipe shall be black iron and pipe fittings shall be malleable iron. Unions shall be installed, where required, for disassembly of piping, valves, and accessories. On the discharge line of the fuel oil pump, a tee-off line with stop valve shall be provided for refilling the fuel tanks of auxiliary generator sets.

3.14.2.3 Fuel control device. A control device shall be installed in the fuel supply system to the burner in a location accessible to the operator. The control device shall provide an infinitely variable range of adjustment and shall maintain a steady flow of fuel at any setting. The control device shall include a calibrated dial and indicating pointer to facilitate burner operating procedures. A gate type, shutoff valve shall be installed in the fuel supply piping between the control device and the fuel pump. The gage valve shall be of the same size as the piping in which it is installed. A relief valve shall be provided and installed in the fuel supply line to the burner. The relief valve shall have a capacity not less than the capacity of the fuel pump. A fuel tank return line shall be connected to the discharge of the relief valve. All low points in the fuel oil system shall be provided with drains to permit complete drainage of the piping.

3.14.2.4 Fuel pump. The fuel pump shall be of the positive displacement type for pumping the required fuels to supply the drier heat demand. This capacity shall be developed when pumping against a discharge pressure not less than the combined friction loss created by the burner and fuel supply system. The pump shall be self-priming, with an electric motor drive, and shall be permanently mounted on the drier frame.

3.14.2.5 Strainer. An oil strainer shall be provided and installed in the fuel supply line in such a manner as to permit easy accessibility for cleaning. The strainer shall have the capacity to remove all foreign matter which might foul the burner.

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3.14.2.6 Advance temperature detector. An advance temperature detecting device shall be provided for compensating the sudden surge of fuel temperature relationship demand in the drier, when charged with large load of aggregate with higher or lower than normal moisture content. The device shall be capable of overriding the set fuel control until temperatures are stabilized.

3.14.2.7 Pressure gage. An oil pressure gage shall be installed in the burner fuel supply system. The gage shall operate with the fuel oils specified for the fuel pump, shall have a heavy-gage metal case and an indicating dial with a range of 0 to 100 lb per square inch gage, and shall be provided with a shutoff cock or valve. The dial shall be graduated in increments of not less than 5 lb per square inch (psi).

3.14.2.8 Combustion chamber and ignition cone. The combustion chamber of the aggregate drier shall consist of a steel cylinder, lined with firebrick conforming to ASTM Designation C27, refractory conforming to ASTM Designation C213, or plastic refractory with equal rating. The ignition cone shall consist of a steel refractory lined cone.

3.14.2.9 Temperature indicator. A temperature indicator shall be provided to indicate the temperature of the aggregate as it is discharged from the drier. The indicator shall have a range of 100°F to 500°F.

3.14.2.10 Driving mechanism. All the necessary gearboxes, shafting, motor, chains, bearings, and any components required for a complete operating drive system shall be provided.

* 3.14.2.11 Gearboxes. The main drive shall be provided with a gearbox to the power source. Gearboxes shall be provided with oil seals to prevent loss of oil and entry of foreign material.

3.14.2.12 Forced draft blower. A centrifugal type blower shall be provided to supply air for atomization to the drier burner. The forced draft blower shall have a centrifugal type wheel, steel shaft and housing, and drive. The forced draft blower shall have a capacity to provide a sufficient quantity of air to the burner to assure efficient operation of the burner at maximum burner capacity. The forced draft blower shall be provided with antifriction type bearings and the intake shall be provided with an expanded metal or wire mesh screen.

3.14.3 Discharge elevator and discharge chute. Means shall be provided at the burner end of the drier drum to discharge the dried aggregate into a discharge chute which will, in turn, direct the aggregate into the boot of a hot-aggregate elevator. The elevator discharge shall be of such a height as to eliminate the need for a hot-aggregate elevator pit.

3.15 Fines collector. The fines collector shall consist of a motor, an exhaust fan, steel stack, centrifugal type cyclone collector(s), screw conveyor, metal duct, all necessary drives, gearboxes, controls, instruments, accessories, and other components necessary for the operation of the unit.

3.15.1 Exhaust fan. The exhaust fan shall be of a type for the intended application and shall provide sufficient draft through the drier and fines collector to result in an efficient separator operation. The exhaust fan

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shall have a single inlet to which the duct from the fines collector shall be attached and a flanged discharge for connection of the stack. The fan housing shall be provided with inspection and cleanout plates.

3.15.2 Stack. A removable stack, of a size compatible with the exhaust fan, shall be provided and shall be flange mounted on the discharge of the exhaust fan. The stack shall be constructed on sheet steel.

3.15.3 Collector. The collector shall be of the centrifugal cyclone type fabricated of welded sheet steel, cast steel, or cast iron, and shall be complete with all necessary flanges for inlet and outlet connections. The capacity shall be such as to be compatible with the exhaust fan.

3.15.4 Collector conveyor. A screw or enclosed belt conveyor shall be attached to the bottom of the fines collector and shall convey the fines from the collector to the boot of the hot-aggregate elevator. The conveyor tube shall be provided with inspection plates and cleanouts and with a discharge opening at the end. The conveyor may be constructed in sections securely bolted together.

3.15.5 Duct work. All metal duct work and fittings required for connecting the exhaust gas outlet of the drier to the inlet of the fines collector, and to connect the outlet from the fines collector to the exhaust fan shall be provided. The duct work shall be fabricated of steel with adequate supporting brackets and transitions. All necessary draw band, gaskets, and flanged connections shall provide a tight duct system, without excess stresses, when assembled for operation. Ducts shall be provided with structural steel supports or shall be self-supporting to assure a sturdy installation. A damper shall be provided for manually regulating the draft.

3.16 Dust collector (bag house). When specified (see 6.2), a fabric filter type dust collector system shall be furnished. The system shall be designed to be used in conjunction with the plant fines collector system and shall be capable of meeting all Federal Environmental Protection Agency Requirements for Emission Control. The collector shall be the manufacturer's standard commercial design and shall be siezed for the capacity of the plant specified. The collector shall be completely assembled, and shall include all components and accessoreis normally furnished with the system including, but not limited, to the following:

a. Collector bag housing with bags.
b. Dust collection hoppers and screws.
c. Inlet and exhaust system.
d. Compressed air system.
e. Electric power system.
f. Safety and operating controls.

3.17 Hot-aggregate elevator. A hot-aggregate elevator of the centrifugal discharged bucket type, totally enclosed, shall be provided. The elevator shall be compatible in height with the upper batch tower unit, and shall have a capacity equal to, or greater than, that of the vibrating screen unit. The elevator shall be totally enclosed, and provided with intake ports.

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3.18 Gradation control unit. The gradation control unit shall consist of a vibrating screen unit with motor and vibrating mechanism, a four-compartment bin of adequate capacity, for the plant, with control gates, bin level indicators, chutes, housings, ladders, platforms, guards, and all other components required for the operation of the unit as specified herein.

3.18.1 Capacity. The gradation control unit shall have a capacity to screen and feed not less than 50 tph of the dried limestone or gravel aggregates.

3.18.2 Vibrating screen. A horizontal or inclined vibrating screen assembly, with interchangeable wire screen cloths of the sizes specified herein, shall be provided. The vibrating screen assembly shall grade the aggregate into four uniform sizes and scalp off oversize material. The screen shall discharge each of the four sizes of aggregate into a separate compartment with the smallest aggregate being discharged into the largest (fine sand) compartment. The top deck shall be provided with an overflow chute to scalp off oversize aggregate. The screen shall be equipped with screen cloths of the following sizes: 2-, 1-, 3/4-, 1/2-, 3/8-, 1/4-, No. 4, and No. 10, all with square openings. The screen shall consist of a stationary frame upon which is mounted a frame with the specified cloths and with a vibrating mechanism. Three screen cloths, of sizes specified for each frame shall be provided with the vibrating screens.

3.18.2.1 Stationary frame. The stationary frame of the screen shall be contructed of structural steel members, or steel castings cross braced to prevent distortion under maximum loads, and operating conditions as specified herein.

3.18.2.2 Live frame. The live frame of the screen assembly shall be constructed of steel plates or structural steel members of rigid construction, with sufficient crossmembers to securely hold the width to insure fitting of screen cloths. The live frame shall be provided with means for holding the replaceable screen cloth securely in place to prevent movement of the cloth during full-load operation. A series of longitudinal bars shall be provided to support the screen cloth. Provisions shall be made for changing cloth by using common tools. The frame shall be balanced to provide an even flow of material across the full width of the cloth.

3.18.2.3 Screen cloth. All screen cloth shall be made of crimping grade, oil-tempered wire, carbon range 0.55 to 0.70, drawn and hardened. Size of wire for 3/8-inch screen openings and above shall be of the standard medium type; for 1/4-inch screen opening and below shall be standard light type. Cloth size opening shall be equal to U.S. Test Sieve of the mesh sizes specified.

3.18.2.4 Vibrating mechanism. A mechanical vibrating mechanism shall be provided for activating the live frame assembly. If material from the deck or decks of the screen impinges on the activating mechanism, the mechanism shall be equipped with a replaceable, protective cover.

3.18.2.5 Screen housing. The entire screen assembly shall be housed with heavy-gage sheet steel to retain dust and heat. Access openings and covers shall be provided at all points requiring inspection or maintenance.

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3.18.3 Bin. The bin shall be fabricated of heavy-gage sheet steel braced and reinforced with structural steel shapes. The bin shall be divided into four compartments capable of meeting the design mix at the plant capacity. Sampling device shall be provided for each bin compartment.

3.18.3.1 Telltales and overflows. Each of the four compartments shall be provided with an overflow chute and telltales to indicate and audibly warn when the level of aggregate within the bin is insufficient for mixer operation. The audible warn may be a horn, bell, or similar device.

3.18.3.2 Sampling. Means shall be provided for obtaining representative samples from each bin compartment. All accessories necessary for obtaining samples shall be furnished.

3.18.3.3 Walkways and ladders. Steel walkways shall be provided at the top of the bin for access to bin compartments and screens for inspection, servicing, and maintenance. The walkways shall be provided with safety tread, handrails, and ladders.

3.18.3.4 Fines feeder. The fines feeder shall consist of a hopper, screw conveyor, drives, supporting frame, and all accessories or other components required for the operation of the fines feeder as specified herein. The fines feeder shall be either ground mounted or integrally mounted with other components.

3.19 Batch unit. The batch unit shall consist of a hopper, transfer pump, batching system, spray bar or spreader bucket, pugmill mixer, discharge chutes, and all other accessories, piping, and components necessary for the operation of the unit. The batch unit shall be supported to provide a minimum height of 10 feet between the discharge gate and ground level.

3.19.1 Capacity. The batch unit shall have a capacity of mixing not less than 2,000 lb per batch.

3.19.2 Transfer pump. The transfer pump shall be the positive displacement type, equipped with a pressure regulating device, all necessary piping, and all-iron body jacketed with hot oil for pumping the bitumen to the batching system from an external source. The pump shall have a capacity compatible to the plant capacity for transferring the bitumen of penetration grade 85 to 100, heated to a temperature of 250°F to 325°F. Penetration grades shall be as determined by ASTM Designation D5. The pump shall be provided with control located on the operator's station.

3.19.3 Measuring and injection system. A measuring and batching system of the volumetric or weight bucket type shall be provided to supply bitumen to the spray bars or spreader bucket. The system shall have an accuracy of within ±1 percent, when using bitumen with a penetration grade of 85 to 100, at a temperature between 250°F and 325°F.

3.19.4 Pugmill mixer. The pugmill mixer shall consist of an enclosed steel chamber in which is mounted not less than two paddle shafts equipped with paddles for mixing aggregate, and bitumen or water into a homogeneous mixture. The size of the pugmill mixer shall be such as to produce the capacity specified in 3.19.1. Paddle arm arrangement shall be such that any

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one paddle or wearing sleeve arm may be removed or replaced without disturbing any of the other paddle arms. The shafts shall be constructed of steel and the paddle arms of high carbon or alloy steel. The paddle tips shall be replaceable, or be covered with a replaceable wearing sleeve, and be constructed of abrasive-resisting cast iron or steel. The pugmill mixer shall be lined with replaceable abrasive-resistant liner. The pugmill mixer shall be jacketed for heating with hot oil. The paddle shafts shall be supported at each end with steel supports and antifriction bearings located outside of the mixing chamber. The pugmill housing shall be constructed of steel and shall have a steel inspection cover. The pugmill shall be provided with the necessary drives or gear trains to receive power from the motor for driving the mixer. The mixer shall be provided with control and timers located at the operator's station. Two adjustable timers shall be provided; one for wet mix, and one for dry mix. Mixer discharge gate shall be either hydraulically or pneumatically operated for unloading.

3.19.5 Controls and instrument panel. All controls, such as weighing scales, thermometers, gages, indicating light signals, levers, drier burner controls, and other accessories for the operation of mixer and gradation screening units, shall be located on the operator's station.

3.20 Automatic batching control system. When specified (see 6.2), an automatic batching control system shall be furnished. The system shall include dry and wet mix timers and automatic cycling device. Provisions shall be made for switching the system to manual operation.

3.21 Piping and accessories. All piping and accessories, which are required for interconnecting of components units to form a complete operating plant, shall be furnished by the contractor in accordance with the arrangement plan of the plant approved by the Government. Where sectionalization is required on the hot oil asphalt lines for assembling and disassembling of the plant, flange connections with fiber gaskets shall be installed. Wire braided flexible hoses shall be provided for the hot oil and asphalt lines for connecting to the hot oil heaters, melters, asphalt storage transfer pump, and the asphalt storage tanks for the pugmill mixer, if required. A jacketed recirculating hot oil asphalt line from the pugmill mixer to the 10,000 gallon asphalt tank shall be provided for normal plant layout. All piping, fittings, valves, and other accessories to make up a complete plant shall be of steel, iron body, or malleable iron. Drier burner fuel piping shall be capable of withstanding 100 psi hydrostatic test pressure; hot oil piping, including jackets, 100 psi; asphalt piping, 50 psi.

3.22 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The units shall be lubricated as specified in Section 5 prior to delivery.

3.23 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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* 3.24 Lifting and tying down attachments. When specified (see 6.2), the component units of the plant shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to MIL-STD-209, type III. Tying down attachments may be identified by stenciling or other suitable marking on the component units. Tying down marking shall clearly indicate that the attachments are intended for tying down of the component units on the carrier when shipped.

* 3.25 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants, such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practicable, after cleaning, and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treatment. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Unless otherwise specified (see 6.2) the color of the finish coat shall be green No. 14064, conforming to FED STD No. 595.

3.26 Data plates.

3.26.1 Instruction plates. The component units which require operational instructions shall be equipped with instruction plates, including warnings and cautions suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the plant, including complete lubricating instructions. All controls shall be identified indicating the direction of motion and the operation to be performed. This may be accomplished by the use of one or more plates. Plates shall be manufactured of noncorrosive durable material, and shall be permanently secured on the plant.

* 3.26.2 Nameplate. A nameplate will be furnished by the contracting officer for each component unit. The contractor shall stamp all necessary data in the blank spaces provided for that purpose, and securely affix it to each unit in a conspicuous place with non-ferrous screws, bolts or rivets not less than 1/8 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

3.27 Special maintenance tools. Such special tools as are required for field maintenance and operation of the equipment shall be packed in the toolbox and shipped with the equipment. Special tools are defined as those not normally included in mechanic's tool kit, but which are designed specifically for a certain purpose on a specific piece of equipment. Means shall be provided to carry those tools and accessories required to operate the unit. Toolboxes, when furnished to hold tools and accessories, shall be made of sheet steel 0.0747-inch (U.S. revised standard gage No. 14) nominal thickness.

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* 3.28 Workmanship.

3.28.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.28.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.28.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each plant shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.2.1 First article inspection. First article inspection shall be performed on one plant when a first article sample is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.2 and the test of 4.3. The first article may be a standard production item from the current inventory provided the plant meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining plants to be furnished under the contract.

4.3 Tests. The component units shall be arranged in the order as shown in the approved arrangement plan of the plant, and the piping and accessories shall be connected to the component units for compatibility and subjected to tests in accordance with 4.3.1 through 4.3.5. Nonconformance to the requirements as specified in Section 3 shall constitute failure and cause for rejection.

4.3.1 Test conditions. Prior to being tested, the plant shall be lubricated. Oils shall be those designated for use in the ambient temperature at which the tests are being performed. During all plant tests, the motors shall be run at normal operating speeds. Where individual components are tested with the use of power sources other than the plant power supply, the component will be operated at the normal plant operating speeds. As services and adjustments are performed, the plant shall be checked for ease of maintenance.

4.3.2 Compatibility. Assemble and operate all the component units in the arrangement as shown in the approved arrangement plan. The test may be conducted with load and without load for at least 1/2 hour each, provided that operating with or without loads will cause no damage to any unit or to the plant. All component units shall assemble and operate compatibly without modifications. The piping, hose, valves, fittings, and accessories shall be assembled, and matchmarks shall be painted on each of the piping, fittings, valves, and accessories, before disassembled for shipment, for the purpose of easy identification when reassembled in the field.

4.3.3 Drying test. Unless otherwise proven by previous test records or a certificate of acceptance for the drying capacity, the drier shall be tested to demonstrate the 50 tph capacity of drying aggregate having 5 percent free moisture.

* 4.3.4 Vehicular component units performance test. The component units equipped with axles and rubber tired wheels shall be towed by a wheeled tractor or truck tractor, over unpaved roads, for not less than 10 miles, then operate cross-country for not less than 1/2 mile. During or after the test on each trailer, trailing, turning, and braking ability shall be observed.

4.3.5 Lifting and tying down attachment tests. When lifting and tying down attachments are required, they shall be tested to conform to 3.24.

4.4 Packaging inspection. The preservation, packing, and marking of the plant shall be inspected to verify conformance to the requirements of Section 5.

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5. PACKAGING

5.1 Preservation. The preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss, and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Large sections and components shall be matchmarked with lusterless white enamel letters. Matchmarking shall be overcoated with clear spar varnish conforming to TT-V-121. Small parts that are removed shall have matchmarking information marked on weatherproof tags attached to mating parts. Waterproof ink shall be used for marking of information on tags.

5.1.1.2 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.3 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.4.2 Machined surfaces. Exposed machined surfaces, shall be coated with type P-6 or P-11 preservative and wrapped or covered as applicable, with barrier material conforming to MIL-B-121, type 1, grade A, class 2 or MIL-B-22191, type 11. The material shall be secured in place with waterproof tape.

5.1.1.5 Gears.

5.1.1.5.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative, or a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.5.2 Enclosed gears. Gear housings containing lubricant conforming to the applicable lubrication order shall be checked for operating level and the gears actuated to insure coating of all interior surfaces.

5.1.1.6 Drive chains.

5.1.1.6.1 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified for enclosed gears.

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5.1.1.6.2 Exposed chains. Exposed chains shall be coated with enough P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chain and sprockets shall be coated with type P-1 preservative.

5.1.1.7 Brakes.

5.1.1.7.1 Brake drum assemblies. All metal surfaces enclosed within brake drums and the braking surface or face of the brake drums shall be coated with a thin film of primer conforming to TT-P-664 or MIL-P-46093. Care shall be exercised to prevent the primer from coming in contact with brake lining, rubber, and rubber impregnated parts.

5.1.1.7.2 Air actuated brake systems. The air actuated brake systems shall be drained of condensation. Interior surfaces of the air supply tanks shall be fogged with type P-9 or P-10 preservative. Excess preservative shall be allowed to drain and the drain plugs reinstalled after preserving the threads with P-10. Air line filters shall be drained and closed. The exhaust ports of relay emergency quick release and relay valves not equipped with exhaust check valves shall be closed by inserting pipe plugs or by sealing the ports with waterproof tape. Weatherproof tags shall be attached to the valves and service line indicating that: "The plugs or tape must be removed from the exhaust ports before placing the equipment in operation." Waterproof ink shall be used for marking of information on tags.

5.1.1.8 Hydraulic systems. The hydraulic fluid supply tank shall be filled to the operating level with approved hydraulic fluid. Where practicable, pistons shall be in their fully retracted position, with any remaining unpainted surfaces coated with type P-1 preservative. Where pistons cannot be fully retracted, exposed areas of normally retractable surfaces shall be coated with type P-6 or P-11 preservative. The type P-6 or P-11 preservative coated surfaces shall be spirally wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2. The wraps shall be secured in place with waterproof tape. All other unpainted exterior surfaces shall be coated with type P-1 preservative. All assemblies actuated by the hydraulic system, shall be secured to prevent movement.

5.1.1.9 Conveyor belts. Metal belt fasteners shall be coated with water resisting varnish conforming to TT-V-119 or MIL-V-13811. The belts on the wheel mounted conveyor shall be removed, belts on other units shall be removed or released from tension. Removed belts shall be compactly rolled. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required to hold belt in compact roll, flat steel strapping conforming to QQ-S-781, class 1, type 1, fininish B, size as applicable, shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping, and the strapping shall not be tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier conforming to PPP-B-1055, class E-2, or H-1, with all laps, seams, and folds sealed with a waterproof adhesive or tape.

5.1.1.10 V-belts; flat belts, and pulleys. Unpainted surfaces of pulleys shall be coated with primer conforming to TT-P-664 or MIL-P-46093. The belts for each unit shall be preserved method 1C-1 or 1C-3. Large flat belts shall be preserved and packaged as specified for conveyor belts.

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5.1.1.11 Drum and rotary elevators. All exterior unpainted metal of the drier drum, rotary elevator, thrust rollers, sprockets, and unpainted metal surfaces within the drum and rotary elevators, shall be coated with type P-1 preservative. Buckets and carrying chains which are subject to movement and damage to the housing of the elevator during shipment shall be removed. The chain shall be coiled and secured with evenly spaced wire ties. The removed buckets and chains shall be consolidated in boxes as specified in 5.1.1.24.

5.1.1.12 Pumps and burners. The interior surfaces of the pump, piping, valves, jets, and nozzles and burners shall be coated with type P-10 preservative in a manner to insure thorough coating of all interior surfaces. Excess preservative shall be drained. All openings shall be sealed with metal or plastic caps or plugs, or sealed with waterproof tape.

5.1.1.13 Fuel tanks. The interior surfaces of fuel tanks, if uncoated or not fabricated of corrosion-resistant material, shall be fogged with type P-10 preservative. Excess preservative shall be drained and the threads of drain plugs shall be coated with P-10 before reinstalling.

5.1.1.14 Pugmill bunker, feeder, dust collector, and shaker. Exterior unpainted ferrous metal surfaces of the units shall be coated with type P-1 preservative. Interior unpainted ferrous metal surfaces of the mixing chamber, bunkers, including shafts, screens, and paddles, shall be coated with type P-1 preservative.

5.1.1.15 Electrical components. Electrical motors shall be preserved level A by the alternate requirements of MIL-E-16298. Other electrical components shall be preserved level A in accordance with MIL-E-17555.

5.1.1.16 Air compressors. The air compressors, parts, and accessories shall be preserved in accordance with MIL-C-3600, level A.

5.1.1.17 Wire rope. The wire rope shall be coated with a preservative conforming to VV-L-751, grade 2 or 3 and secured to prevent movement.

5.1.1.18 Openings. Openings in the equipment which will trap free falling water shall be sealed by any suitable means to prevent the entry of water.

5.1.1.19 Piping equipment. Loose piping equipment required to complete the water-supply system shall be preserved method I or III as applicable. Exposed threaded surfaces shall be protected against damage by the use of metal or plastic thread protectors.

5.1.1.20 Lubrication. Lubricate all grease fittings with type P-11 preservative. Wheel bearings should be lubricated with compound specified by the manufacturer. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-9.

5.1.1.21 Rubber tires. Tires shall be inflated to 10 lb above the specified operating pressure.

5.1.1.22 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

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5.1.1.23 Technical publications. Technical publications for each piece of equipment shall be preserved in accordance with MIL-P-116, method 1C-1 or 1C-3.

5.1.1.24 Consolidation. The dash compartment and tool boxes shall be utilized for the consolidation of loose components, tools, and publications. The remainder of the components that will not fit in the dash compartment or tool boxes shall be consolidated, along with disassembled components and repair parts, in boxes conforming to PPP-B-601, military overseas type or PPP-B-621, class 2. Boxes shall be placed in the cab or secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Level C. The complete asphalt plant shall be preserved and packaged in a manner that will provide protection against deterioration and damage from the contractor to the initial destination. This level may conform to the contractor's standard practice when such meets the requirements of the level.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Trailer mounted units (except wheel mounted conveyors). Trailer mounted units shall be shipped mobile and uncrated. Movable components shall be blocked and braced as needed to prevent movement during shipment.

5.2.1.2 Wheel mounted conveyors. The conveyor shall be disassembled. The conveyor belt and electric motor shall be packed in boxes conforming to PPP-B-601, military overeas type or PPP-B-621, class 2. The wheels and chassis components shall be packed in open type crates conforming to MIL-C-52950, style A or B, type optional. Sections of the conveyor shall be shipped uncrated or bundled. When bundling is used the sections shall be nested, arranged, and secured with suitable wood blocking and battens, as required, to form compact nonshifting bundles. Strapping shall be 0.035 by 1-1/4 inch flat steel conforming to QQ-S-781, class 1, type 1, finish B. The bottom of the secured load shall be fitted with wooden skids extending the full width of the load to accommodate forks and slings for handling.

5.2.1.3 Components not trailer mounted. Components not trailer mounted shall be packed in accordance with the level A requirements of MIL-STD-794. Containers shall be selected from table II of MIL-STD-794 for level A requirements.

5.2.1.4 Piping and accessories. Piping suitable for bundling shall be bundled in secured lifts. The piping shall be nested, arranged, and secured with bolts or steel straps, or a combination of both, to form compact nonshifting bundles. Suitable wood blocking shall be used, as required, under strapping to eliminate large voids or irregular shaped bundles. Strapping shall be 0.023 by 3/4-inch flat steel strapping conforming to QQ-S-781, class 1, type I, finish B. Strapping shall be spaced not to exceed approximately 36 inches on center, with end straps not more than 18 inches from each end. Metal edge protectors shall be used under strapping when strapping bears on the pipe. The bundles shall be packed in open type crates conforming to MIL-C-52950, style A or B, type V. Short pieces of pipe, hose fittings, and

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accessories shall be packed in close-fitting boxes conforming to PPP-B-621, class 2 or PPP-B-601, military overseas type. The contents in the boxes and crates shall be cushioned, blocked, and braced to prevent movement.

5.2.2 Level B. Level B requirements shall be the same as level A except that components not trailer mounted shall be packed in containers designated for level B from table II of MIL-STD-794.

5.2.3 Level C. The complete plant shall be prepared for shipment in a manner which will insure arrival at destination in satisfactory condition. Loose components that are subject to damage during shipment shall be blocked, braced, or otherwise secured to prevent movement and damage during shipment. The toolbox lids shall be closed and secured in a manner to prevent pilferage. When suffficient space is available, packaged and unpackaged components required for each unit or section shall be placed and secured on the unit or sections for which intended in any available space that will not interfere with lifting or mobility of the units. Boxed and crated components shall be shipped at the same time as the mobility units. Packing and containers shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This completely equipped plant, batch type, electric-motor-driven, of not less than 50 tph capacity, is intended primarily for use in asphaltic paving of airfield, road, street, parking lot, large floor area surfaces, or other uses.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When specifications and standards shall be other than as specified (see 2.1.1).
c. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.5).
d. When dust collector (baghouse) is required (see 3.16).
e. When automatic batching control system shall be furnished (see 3.20).
f. When fungus resistance is required (see 3.23).
g. When lifting and tying down attachments are required (see 3.24).
h. Color of finish coat required if other than specified (see 3.25).
i. Level of preservation and level of packing required (see 5.1 and 5.2).

* 6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.4 Electrical power requirements. The electrical power requirements of the plant is compatible with the electrical power available from tactical utility generator sets in the DoD Mobile Electrical Power Engine Generator Set Family. After power requirements of the plant has been determined, the model(s) of generator set(s) required may be selected from MIL-STD-633, and obtained in accordance with applicable departmental policy.

6.5 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete plant unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-29160A(YD) MIXING PLANT, ASPHALT, BATCH TYPE, PORTABLE, ELECTRIC-MOTOR-DRIVEN, 50 TON PER HOUR CAPACITY

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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